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Device integration schemes leveraging a bulk semiconductor substrate having a <111> crystal orientation

Abstract

Structures including devices, such as transistors, integrated on a bulk semiconductor substrate and methods of forming a structure including devices, such as transistors, integrated on a bulk semiconductor substrate. The bulk semiconductor substrate contains a single-crystal semiconductor material having a diamond crystal lattice structure and a <111> crystal orientation. A first transistor is formed in a first device region of the bulk semiconductor substrate, and a second transistor is formed in a second device region of the bulk semiconductor substrate. The second transistor includes a layer stack on the bulk semiconductor substrate, and the layer stack includes a layer comprised of a III-V compound semiconductor material.

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Background/Summary

BACKGROUND

(1) The present invention relates to semiconductor device fabrication and integrated circuits and, more specifically, to structures including devices, such as transistors, integrated on a bulk semiconductor substrate and methods of forming a structure including devices, such as transistors, integrated on a bulk semiconductor substrate.

(2) High-voltage power electronic devices, such as high-electron-mobility transistors, may be fabricated using III-V compound semiconductors to exploit their material properties, such as a carrier mobility that is greater than the carrier mobility of silicon. III-V compound semiconductors include Group III elements (aluminum, gallium, indium) combined with Group V elements (nitrogen, phosphorus, arsenic, antimony). A high-electron-mobility transistor may include a heterojunction between crystalline III-V compound semiconductor materials having different band gaps, such as a heterojunction between binary gallium nitride and ternary aluminum-gallium nitride. During operation, a two-dimensional electron gas is formed near an interface at the heterojunction and defines the channel of the high-electron-mobility transistor.

(3) The integration of high-electron-mobility transistors with either field-effect transistors or heterojunction bipolar transistors formed by complementary-metal-oxide-semiconductor (CMOS) processing on the same chip has proven to be a complex process. Integration may be achieved by wafer bonding or through the use of engineered or hybrid substrates, which by their nature lend significant complexity to processes integrating high-electron-mobility transistors with these other types of transistors.

(4) Structures including devices, such as transistors, integrated on a bulk semiconductor substrate and methods of forming a structure including devices, such as transistors, integrated on a bulk semiconductor substrate are needed.

SUMMARY

(5) In an embodiment of the invention, a structure includes a bulk semiconductor substrate comprised of a single-crystal semiconductor material having a diamond crystal lattice structure and a <111> crystal orientation. The bulk semiconductor substrate has a first device region and a second device region. The structure includes a first transistor in the first device region of the bulk semiconductor substrate, and a second transistor in the second device region of the bulk semiconductor substrate. The second transistor includes a layer stack on the bulk semiconductor substrate, and the layer stack includes a layer comprised of a III-V compound semiconductor material.

(6) In an embodiment of the invention, a method includes providing a bulk semiconductor substrate comprised of a single-crystal semiconductor material having a diamond crystal lattice structure and a <111> crystal orientation. The method further includes forming a first transistor in a first device region of the bulk semiconductor substrate, forming a layer stack including a layer comprised of a III-V compound semiconductor material in a second device region of the bulk semiconductor substrate, and forming a second transistor using the layer stack.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various embodiments of the invention and, together with a general description of the invention given above and the detailed description of the embodiments given below, serve to explain the embodiments of the invention. In the drawings, like reference numerals refer to like features in the various views.

(2) FIGS. 1-6 are cross-sectional views of a structure at successive fabrication stages of a processing method in accordance with embodiments of the invention.

(3) FIG. 3A is an enlarged cross-sectional view of a portion of FIG. 3 in accordance with embodiments of the invention.

(4) FIG. 7 is a cross-sectional view of a structure in accordance with an alternative embodiment of the invention.

(5) FIGS. 8 and 9 are cross-sectional views of a structure at successive fabrication stages of a processing method in accordance with an alternative embodiment of the invention.

(6) FIG. 10 is a cross-sectional view of a structure in accordance with an alternative embodiment of the invention.

DETAILED DESCRIPTION

(7) With reference to FIG. 1 and in accordance with embodiments of the invention, a semiconductor substrate **10** is provided that contains a single-crystal semiconductor material, such as single-crystal silicon. The semiconductor substrate **10** has a top surface **12**, which may be planar. The semiconductor substrate **10** is a bulk substrate containing a single-crystal semiconductor material (e.g., single-crystal silicon). In an embodiment, the single-crystal semiconductor material of the semiconductor substrate **10** may have a diamond crystal lattice structure with a $\langle 111 \rangle$ crystal orientation as specified by Miller indices. In an embodiment, the semiconductor substrate **10** may contain single-crystal silicon with a diamond crystal lattice structure having a $\langle 111 \rangle$ crystal orientation. For a semiconductor substrate **10** having a $\langle 111 \rangle$ crystal orientation, the (111) crystallographic plane is parallel to the top surface **12** of the semiconductor substrate **10**, and the crystallographic direction is normal to the (111) plane. The (100) crystallographic axes do not lie in the plane of the top surface **12**. The semiconductor substrate **10** may be characterized as a non-silicon-on-insulator substrate (i.e., a non-SOI) substrate) that lacks a buried oxide layer characteristic of a silicon-on-insulator (SOI) substrate. In an embodiment, the semiconductor substrate **10** may be fully comprised of semiconductor material with a $\langle 111 \rangle$ crystal orientation.

(8) Shallow trench isolation regions **14** are formed that extend from the top surface **12** of the semiconductor substrate **10** into the semiconductor substrate **10**. The shallow trench isolation regions **14** may contain a dielectric material deposited by chemical vapor deposition into trenches etched in the semiconductor substrate **10**, polished, and deglazed. The dielectric material contained in the shallow trench isolation regions **14** may comprise silicon dioxide, silicon nitride, silicon carbide, silicon-rich silicon dioxide, or a combination of two or more of these materials. The shallow trench isolation regions **14** may extend to a depth, d_1 , relative to the top surface **12** into the semiconductor substrate **10**. The shallow trench isolation regions **14** surround and define multiple device regions **16**, **18**, **20**. In an embodiment, the top surface **12** in the device region **18** may be coplanar with the top surface **12** in the device region **16** and also coplanar with the top surface **12** in the device region **20**.

(9) A gate conductor layer **22** comprised of, for example, doped polycrystalline silicon (i.e., doped polysilicon) and a gate dielectric layer **24** comprised of an electrical insulator, such as silicon dioxide, are formed on the semiconductor substrate **10**. These layers **22**, **24** are formed on the top surface **12** in all of the device regions **16**, **18**, **20**.

(10) A hardmask **26** may be formed over the layers **22**, **24** on the semiconductor substrate **10** and patterned to include an opening generally located over the device region **18**. The hardmask **26** may be comprised of a dielectric material, such as silicon nitride, and may be patterned by lithography and etching processes. The section of the gate conductor layer **22** and gate dielectric layer **24** exposed by the opening in the hardmask **26** may be removed by etching with an etching process, such as a reactive ion etching process, which exposes the top surface **12** of the semiconductor substrate **10** in device region **18**.

(11) A trench **30** is subsequently formed in the semiconductor substrate **10** at the location of the opening in the hardmask **26** by etching with another etching process, such as a reactive ion etching process. The trench **30** may extend to a trench bottom located at a surface **32** of the semiconductor substrate **10** and may have side surfaces or sidewalls **29**, **31**. The trench **30** may be surrounded by

the shallow trench isolation regions **14** defining the device region **18**. The surface **32** may be located at a depth, **d2**, in the semiconductor substrate **10** relative to the top surface **12** that is greater than the depth, **d1**, of the shallow trench isolation regions **14**. The hardmask **26** protects the sections of the gate conductor layer **22** and gate dielectric layer **24** in the device region **16** and the device region **20** during the etching processes. In an embodiment, the surface **32** may be planar and lack topography. In an embodiment, the top surface **12** may be planar, the surface **32** may be planar, and the planes of the top surface **12** and the surface **32** may be parallel.

(12) With reference to FIG. **2** in which like reference numerals refer to like features in FIG. **1** and at a subsequent fabrication stage of the processing method, sidewall spacers **34** are formed adjacent to the sidewalls **29**, **31** of the trench **30**. The sidewall spacers **34** may extend from the top surface **12** of the semiconductor substrate **10** to the surface **32** at the bottom of the trench **30**. The sidewall spacers **34** may be formed by depositing a liner layer comprised of a dielectric material (e.g., silicon nitride) and etching the deposited liner layer with an anisotropic etching process, such as a reactive ion etching process.

(13) With reference to FIGS. **3**, **3A** in which like reference numerals refer to like features in FIG. **2** and at a subsequent fabrication stage of the processing method, a layer stack **36** containing one or more compound semiconductor layers is formed on the surface **32** of the semiconductor substrate **10** that is located inside the trench **30**. In an embodiment, the layer stack **36** may include at least one crystalline layer comprised of a III-V compound semiconductor material. In an embodiment, the layer stack **36** may include at least one crystalline layer comprised of a binary III-V compound semiconductor material. In an embodiment, the layer stack **36** may include at least one crystalline layer comprised of a ternary III-V compound semiconductor material. In an embodiment, the layer stack **36** may include multiple crystalline layers comprised of different III-V compound semiconductor materials. In an embodiment, the layer stack **36** may include at least one crystalline layer comprised of a binary III-V compound semiconductor material and at least one crystalline layer comprised of a ternary III-V compound semiconductor material. In an embodiment, the layer stack **36** may include one or more crystalline layers that contain gallium nitride or a ternary III-V compound semiconductor material based on gallium nitride (e.g., aluminum gallium nitride). In an embodiment, the layer stack **36** may include one or more crystalline layers that contain gallium and nitrogen.

(14) The layer stack **36** may be formed by an epitaxial growth process. Each layer of the layer stack **36** may have a crystal structure that is single crystal or, alternatively, a crystal structure that is substantially single crystal with varying levels of crystalline defectivity present. The $\langle 111 \rangle$ crystal orientation of the semiconductor material (e.g., single-crystal silicon) of the semiconductor substrate **10** promotes epitaxial growth of the III-V compound semiconductor material(s), such as gallium nitride, of the layer stack **36** with low crystalline defectivity through closer lattice matching than possible with substrates having a $\langle 100 \rangle$ crystal orientation. In particular, the atoms on the surface plane of the semiconductor substrate **10** with a $\langle 111 \rangle$ crystal orientation are arranged in hexagons, which may lattice match reasonably well with the crystal structure of one or more of the compound semiconductor materials (e.g., the wurtzite crystal structure of gallium nitride that is based on a binary hexagonal close-packed crystal system) in the layer stack **36**.

(15) In an embodiment, the layer stack **36** may be formed by a selective epitaxial growth process in which semiconductor material does not form on dielectric surfaces, such as the hardmask **26** and the sidewall spacers **34**. In an embodiment, the layer stack **36** may be formed by a non-selective epitaxial growth process in which semiconductor material is deposited and patterned by lithography and etching processes. In an embodiment, the sidewalls **37** of the layer stack **36** may be located adjacent to, and spaced from, the trench sidewalls **29**, **31** and, in that instance, the trench **30** may be substantially filled by the layer stack **36**. In the representative embodiment, the layer stack **36** has sidewalls **37** that are inwardly inclined away from the trench sidewalls **29**, **31** to define, for example, a trapezoidal shape and to space the sidewalls **37** from the trench sidewalls **29**, **31**. In an

embodiment, the sidewalls **37** may converge at a top surface **35** of the layer stack **36** that may be coplanar or substantially coplanar with the top surface **12** of the semiconductor substrate **10**. Isolation regions (not shown) may be formed at the top surface **35** of the layer stack **36** by, for example, a masked implantation of either nitrogen or argon.

(16) In an embodiment and as best shown in FIG. 3A, the layer stack **36** may include a buffer layer **76**, a channel layer **78**, a spacer layer **80**, and a barrier layer **82**. The layers **76**, **78**, **80**, **82** may be serially formed using an epitaxial growth process, such as metalorganic chemical vapor deposition. The layers **76**, **78**, **80**, **82** may each have a crystal structure that is single crystal or, alternatively, a crystal structure that is substantially single crystal with varying levels of crystalline defectivity present. One or more of the layers **76**, **78**, **80**, **82** may include multiple sub-layers characterized by varying composition or doping. The buffer layer **76** may contain a III-V compound semiconductor material, such as gallium nitride, that is tailored in terms of material composition, doping, and/or layer thickness to accommodate lattice mismatch between the material of the semiconductor substrate **10** and the material of the channel layer **78**. The channel layer **78**, which is disposed over the buffer layer **76**, may contain a III-V compound semiconductor material, such as gallium nitride. The spacer layer **80** and the barrier layer **82** are disposed over the channel layer **78** with the spacer layer **80** between the channel layer **78** and the barrier layer **82**. The spacer layer **80** may be thin and may contain a III-V compound semiconductor, such as aluminum nitride. The barrier layer **82** may contain a III-V compound semiconductor, such as aluminum gallium nitride, aluminum nitride or indium aluminum nitride, that has a heterogeneous interface with the channel layer **78** of different composition. The spacer layer **80** and barrier layer **82**, along with the material properties of the channel layer **78**, contribute to creating a two-dimensional electron gas, during device operation, at the heterogeneous interface that is filled with highly-mobile and abundant electrons.

(17) With reference to FIG. 4 in which like reference numerals refer to like features in FIG. 3 and at a subsequent fabrication stage of the processing method, a dielectric layer **38** may be deposited and patterned by photolithography and etching processes such that the layer stack **36** in the device region **18** is covered. The hardmask **26** is removed from the device region **16** and the device region **20** by an etching process, after the dielectric layer **38** is deposited and patterned, to expose the gate conductor layer **22** and gate dielectric layer **24**.

(18) The gate conductor layer **22** and gate dielectric layer **24** may be subsequently patterned by photolithography and etching processes to define a gate structure of a field-effect transistor **42** in the device region **16**. The gate conductor layer **22** and gate dielectric layer **24** are removed from the device region **20** during patterning. Additional elements of the field-effect transistor **42** may be fabricated by complementary-metal-oxide (CMOS) processing to form a device structure in the device region **16** of the semiconductor substrate **10**. The field-effect transistor **42** may additionally include source/drain regions **40**, and a channel region beneath the gate structure. The channel region and the source/drain regions **40** of the field-effect transistor **42** contain respective portions of the single-crystal semiconductor material of the semiconductor substrate **10**. The source/drain regions **40** are positioned in the semiconductor substrate **12** at least in part beneath the top surface **12**, and the channel region is positioned beneath the top surface **12** generally between the source/drain regions **40**. The patterned gate conductor layer **22** and gate dielectric layer **24** contained in the gate structure of the field-effect transistor **42** may be positioned as device layers on and above the top surface **12** of the semiconductor substrate **10**. In an embodiment, the patterned gate dielectric layer **24** contained in the gate structure of the field-effect transistor **42** may be positioned as a device layer directly on the top surface **12** of the semiconductor substrate **10**.

(19) A bipolar junction transistor **44** may be fabricated as a device structure in the device region **20** of the semiconductor substrate **10**. The bipolar junction transistor **44** may include multiple terminals in the form of a collector **46** positioned in the semiconductor substrate **10**, an emitter **48**, and a base layer **50** arranged between the collector **46** and emitter **48**. In alternative embodiments, a collector-up bipolar junction transistor may be formed with the emitter arranged as the terminal in

the semiconductor substrate **10**. The emitter **48** and the base layer **50** may be positioned as device layers on and above the top surface **12** of the semiconductor substrate **10**. In an embodiment, the base layer **50** may be positioned as a device layer directly on the top surface **12** of the semiconductor substrate **10**. The collector **46**, which contains a portion of the semiconductor material of the semiconductor substrate **10**, may be positioned in the semiconductor substrate **10** at least in part, and preferably fully, beneath the top surface **12**. The base layer **50** may contain single-crystal semiconductor material (e.g., silicon-germanium) that is epitaxially grown on the top surface **12** of the semiconductor substrate **10**. In an embodiment, the collector **46** and emitter **48** may contain n-type semiconductor materials and the base layer **50** may contain p-type semiconductor material to define an NPN transistor. The bipolar junction transistor **44** may be formed by bipolar-complementary-metal-oxide (BiCMOS) processing, which is variant of CMOS processing.

(20) The field-effect transistor **42** and the bipolar junction transistor **44** constitute different types or classifications of transistor structures. A difference between the field-effect transistor **42** and the bipolar junction transistor **44** is that only majority charge carriers flow in the field-effect transistor **42**, whereas both majority and minority charge carriers flow in the bipolar junction transistor **44**. The field-effect transistor **42** and the bipolar junction transistor **44** do not include any layers of silicon carbide in their respective constructions and, therefore, are free of silicon carbide. The field-effect transistor **42** and the bipolar junction transistor **44** are both formed on semiconductor material characterized by the same $\langle 111 \rangle$ crystal orientation as the semiconductor material used to form the layer stack **36**.

(21) With reference to FIG. 5 in which like reference numerals refer to like features in FIG. 4 and at a subsequent fabrication stage of the processing method, a transistor **54** is formed as a device structure using the layer stack **36**. To that end, the dielectric layer **38** is patterned by photolithography and etching processes in device region **18** to define an opening in which a gate electrode **52** of the transistor **54** is formed. The gate electrode **52** may be comprised of a metal, such as a metal nitride, that is deposited and patterned with lithography and etching processes to define a given shape. A source region **56** and a drain region **58** of the transistor **54** may be formed by patterning openings in the dielectric layer **38** by lithography and etching processes, and then forming a metal, such as a metal nitride, in the patterned openings. Metal atoms from the source region **56** and drain region **58** may diffuse into the layer stack **36**.

(22) The transistor **54** is not formed by CMOS processes and, therefore, may be considered to be a non-CMOS transistor. In an embodiment, the transistor **54** may be a high-electron-mobility transistor (HEMT). In an embodiment, the transistor **54** may be a metal-insulator-semiconductor high-electron-mobility transistor (MISHEMT). In an embodiment, the transistor **54** may be a metal-oxide-semiconductor high-electron-mobility transistor (MOSHEMT).

(23) Although the $\langle 111 \rangle$ crystal orientation of the semiconductor material (e.g., single-crystal silicon) of the semiconductor substrate **10** enables the epitaxial growth of the III-V compound semiconductor material(s) of the layer stack **36** with low crystalline defectivity, the field-effect transistor **42** and/or the bipolar junction transistor **44** may exhibit non-optimized electrical performance due to the $\langle 111 \rangle$ crystal orientation in comparison to comparable transistors formed on semiconductor substrates of other crystal orientations. Many of the structural and electronic properties of a single-crystal semiconductor substrate are highly anisotropic and dependent on crystal orientation. Nonetheless, the implementation of the semiconductor substrate **10** with a $\langle 111 \rangle$ crystal orientation permits the integration of the transistor **54** with the field-effect transistor **42** and/or the bipolar junction transistor **44** on the same semiconductor substrate **10** absent a complex manufacturing process, such as wafer bonding, or the use of an engineered or hybrid substrate (e.g., an SOI substrate with one or more crystal orientations for the device layer), which may be considered to be an acceptable trade-off for poorer electrical performance.

(24) With reference to FIG. 6 in which like reference numerals refer to like features in FIG. 5 and

at a subsequent fabrication stage of the processing method, middle-of-line processing and back-end-of-line processing follow, which includes formation of contacts, vias, and wiring for an interconnect structure **60** positioned over the semiconductor substrate **10** and above the transistors **42**, **44**, **54**. Various metallization levels, such as a first metallization (M1) level, of the interconnect structure **60** may be formed that are coupled by the contact level with the field-effect transistor **42**, the bipolar junction transistor **44**, and the transistor **54**. In that regard, the interconnect structure **60** may include one or more dielectric layers **62**, a metallization level having metal lines **66**, and a contact level having contacts **64** that couple the metal lines **66** to the field-effect transistor **42**, the bipolar junction transistor **44**, and the transistor **54**.

(25) With reference to FIG. **7** and in accordance with alternative embodiments, the sidewalls **37** of the layer stack **36** may be coextensive with the trench sidewalls **29**, **31** and, in this instance, the trench **30** may be fully filled by the layer stack **36**. Growth conditions may be selected to impart a given shape to the layer stack **36**. Processing may continue as described in connection with FIGS. **4-6**.

(26) With reference to FIG. **8** and in accordance with alternative embodiments, the trench **30**, the layer stack **36**, and the transistor **54** may be fully formed after the completion of the formation of the field-effect transistor **42** and the bipolar junction transistor **44**. In that regard, the one or more dielectric layers **62** may be formed and patterned to provide an opening **68** defining a pathway to the top surface **12** of the semiconductor substrate **10** in device region **18**. In the representative embodiment, the opening **68** is formed before forming the contacts **64** and metal lines **66**. In an embodiment, the opening **68** may be formed after forming the contacts of the contact level that are coupled to the field-effect transistor **42** and the bipolar junction transistor **44**. In an embodiment, the opening **68** may be formed after forming the metal lines **66** of the first metallization level and the contacts **64** of the contact level coupling the metal lines **66** to the field-effect transistor **42** and the bipolar junction transistor **44**.

(27) With reference to FIG. **9** in which like reference numerals refer to like features in FIG. **8** and at a subsequent fabrication stage of the processing method, the trench **30** is subsequently formed in the semiconductor substrate **10** at the location of the opening **68** in the one or more dielectric layers **62**. The sidewall spacers **34**, the layer stack **36**, and the transistor **54** are subsequently formed using the trench **30** in the semiconductor substrate **10**. The opening **68** may be refilled with dielectric material after forming the transistor **54**, and the interconnect structure **60** may be completed by forming the contacts **64** and metal lines **66** coupled to the transistor **54**.

(28) With reference to FIG. **10** and in accordance with alternative embodiments, the layer stack **36** may be formed on the top surface **12** of the semiconductor substrate **10** without the prior formation of the trench **30**. By forgoing trench formation, the layer stack **36** and the transistor **54** formed using the layer stack **36** may be located in the same plane as the field-effect transistor **42** and the bipolar junction transistor **44**. More specifically, the layer stack **36**, the gate structure of the field-effect transistor **42**, and the emitter **48** and the base layer **50** of the bipolar junction transistor **44** may be positioned on the top surface **12**, which may be considered to provide a common plane for the transistors **42**, **44**, **54**.

(29) The methods as described above are used in the fabrication of integrated circuit chips. The resulting integrated circuit chips can be distributed by the fabricator in raw wafer form (e.g., as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. In the latter case, the chip is mounted in a single chip package (e.g., a plastic carrier, with leads that are affixed to a motherboard or other higher level carrier) or in a multichip package (e.g., a ceramic carrier that has either or both surface interconnections or buried interconnections). In any case, the chip may be integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either an intermediate product or an end product.

(30) References herein to terms modified by language of approximation, such as “about”, “approximately”, and “substantially”, are not to be limited to the precise value specified. The

language of approximation may correspond to the precision of an instrument used to measure the value and, unless otherwise dependent on the precision of the instrument, may indicate $\pm 10\%$ of the stated value(s).

(31) References herein to terms such as “vertical”, “horizontal”, etc. are made by way of example, and not by way of limitation, to establish a frame of reference. The term “horizontal” as used herein is defined as a plane parallel to a conventional plane of a semiconductor substrate, regardless of its actual three-dimensional spatial orientation. The terms “vertical” and “normal” refer to a direction perpendicular to the horizontal, as just defined. The term “lateral” refers to a direction within the horizontal plane.

(32) A feature “connected” or “coupled” to or with another feature may be directly connected or coupled to or with the other feature or, instead, one or more intervening features may be present. A feature may be “directly connected” or “directly coupled” to or with another feature if intervening features are absent. A feature may be “indirectly connected” or “indirectly coupled” to or with another feature if at least one intervening feature is present. A feature “on” or “contacting” another feature may be directly on or in direct contact with the other feature or, instead, one or more intervening features may be present. A feature may be “directly on” or in “direct contact” with another feature if intervening features are absent. A feature may be “indirectly on” or in “indirect contact” with another feature if at least one intervening feature is present.

(33) The descriptions of the various embodiments of the present invention have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A structure comprising: a bulk semiconductor substrate comprising a single-crystal semiconductor material having a $\langle 111 \rangle$ crystal orientation, the bulk semiconductor substrate having a first device region, a second device region, and a trench; a field-effect transistor including a source/drain region positioned in the single-crystal semiconductor material of the first device region of the bulk semiconductor substrate; and a non-CMOS transistor in the second device region of the bulk semiconductor substrate, the non-CMOS transistor including a layer stack inside the trench on the single-crystal semiconductor material of the bulk semiconductor substrate, and the layer stack including a first layer comprising a first III-V compound semiconductor material.
2. The structure of claim 1 wherein the single-crystal semiconductor material is single-crystal silicon, and the first III-V compound semiconductor material is gallium nitride.
3. The structure of claim 1 wherein the bulk semiconductor substrate includes a top surface, and further comprising: a shallow trench isolation region in the bulk semiconductor substrate, the shallow trench isolation region laterally positioned between the first device region and the second device region.
4. The structure of claim 3 wherein the trench extends to a greater depth into the bulk semiconductor substrate relative to the top surface than the shallow trench isolation region.
5. The structure of claim 1 wherein the trench includes a plurality of sidewalls, and further comprising: a spacer on each of the plurality of sidewalls, the spacer comprised of a dielectric material.
6. The structure of claim 1 wherein the trench includes a first plurality of sidewalls, the layer stack includes a second plurality of sidewalls, and the second plurality of sidewalls are located adjacent to the first plurality of sidewalls.

7. The structure of claim 6 wherein the second plurality of sidewalls of the layer stack are inclined inwardly away from the first plurality of sidewalls of the trench.
 8. The structure of claim 6 wherein the second plurality of sidewalls of the layer stack are spaced from the first plurality of sidewalls of the trench.
 9. The structure of claim 1 wherein the trench includes a first plurality of sidewalls, the layer stack includes a second plurality of sidewalls, and the second plurality of sidewalls are coextensive with the first plurality of sidewalls.
 10. The structure of claim 1 further comprising: an interconnect structure positioned over the bulk semiconductor substrate, the interconnect structure including a first plurality of contacts coupled to the field-effect transistor and a second plurality of contacts coupled to the non-CMOS transistor.
 11. The structure of claim 1 further comprising: a shallow trench isolation region in the bulk semiconductor substrate, the shallow trench isolation region laterally positioned between the first device region and the second device region.
 12. The structure of claim 1 wherein the non-CMOS transistor is a high-electron-mobility transistor.
 13. The structure of claim 12 wherein the high-electron-mobility transistor includes a gate electrode, a source region, and a drain region, and the gate electrode, the source region, and the drain region each comprise a metal.
 14. The structure of claim 1 wherein the layer stack includes a second layer comprising a second III-V compound semiconductor material.
 15. The structure of claim 14 wherein the second III-V compound semiconductor material differs from the first III-V compound semiconductor material.
 16. The structure of claim 1 wherein the bulk semiconductor substrate includes a top surface, and the layer stack includes a top surface that is coplanar with the top surface of the bulk semiconductor substrate.
 17. The structure of claim 1 wherein the bulk semiconductor substrate includes a top surface, and the layer stack includes a top surface that is substantially coplanar with the top surface of the bulk semiconductor substrate.
 18. The structure of claim 1 wherein the non-CMOS transistor is a metal-insulator-semiconductor high-electron-mobility transistor or a metal-oxide-semiconductor high-electron-mobility transistor.
 19. A method comprising: forming a trench in a bulk semiconductor substrate comprising a single-crystal semiconductor material having a <111> crystal orientation, wherein the bulk semiconductor substrate includes a first device region and a second device region; forming field-effect transistor including a source/drain region positioned in the single-crystal semiconductor material of the first device region of the bulk semiconductor substrate; forming a layer stack including a layer comprising a III-V compound semiconductor material inside the trench in the second device region of the bulk semiconductor substrate, wherein the second device region comprises the single-crystal semiconductor material; and forming a non-CMOS transistor using the layer stack.
 20. The method of claim 19 further comprising: forming a shallow trench isolation region in the bulk semiconductor substrate, wherein the shallow trench isolation region is laterally positioned between the first device region and the second device region.
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